

Prediction of Monoclonal Antibodies Pharmacokinetics in Human: Identification of a Reference Neonatal Fc Receptor (FcRn) Binding Affinity Using Physiologically Based Pharmacokinetic (PBPK) Modeling

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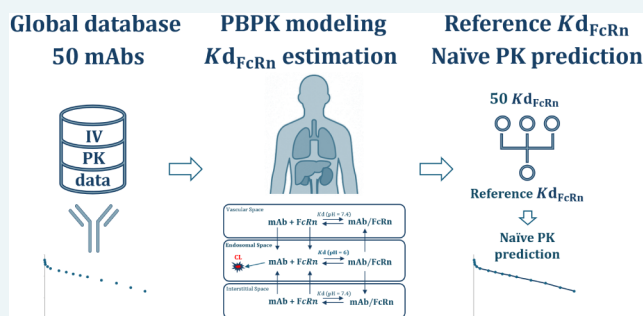
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ABSTRACT: Prediction of monoclonal antibody (mAb) pharmacokinetics (PK) in drug development remains challenging due to the lack of a standardized method for predicting elimination based on mechanistic pathways. Among the processes implemented in the physiologically based pharmacokinetic (PBPK) models for large molecules, FcRn-mediated recycling constitutes the predominant mechanism influencing the elimination of mAbs. In the present study, we assessed the predictivity of a generic value for the dissociation constant (K_d) for FcRn ($K_{d_{FcRn}}$) in humans, identified based on clinical data, to provide means for mechanism-based PK projections for mAbs in first-in-human (FIH) trials. We compiled a database of digitalized linear PK profiles for 50 mAbs administered intravenously in humans. Subsequently, the database was randomly divided into a training and a test data set, using a 7:3 ratio. For each drug in the training data set, a generic PBPK model was set up in PK-Sim, and a drug-specific $K_{d_{FcRn}}$ parameter was estimated through data fitting. The median of estimated drug-specific $K_{d_{FcRn}}$ was $1.05 \mu\text{M}$ and was used for naïve predictions of the PK of the drugs in the test data set. Plasma exposure (AUC) and terminal half-life were accurately predicted for 80% and 60% of the drugs in the test data set, respectively, with a prediction error within the 0.80–1.25-fold range. Additionally, 100% of the test data set showed prediction errors within the 0.50–2.00-fold range for both plasma exposure and half-life. The median of the estimated drug-specific $K_{d_{FcRn}}$ determined using the whole database with 50 mAbs was $1.07 \mu\text{M}$ and was retained after evaluation as a more accurate default $K_{d_{FcRn}}$ value. The reported results provide a large database of mAbs PBPK models with estimated $K_{d_{FcRn}}$ values using PK-Sim, and a validated default $K_{d_{FcRn}}$ value of $1.07 \mu\text{M}$ to perform naïve predictions of mAbs linear PK in the context of FIH trials.

KEYWORDS: monoclonal antibody, $K_{d_{FcRn}}$, PBPK, first-in-human



Antibodies, also referred to as immunoglobulins (Igs), are large proteins utilized by the immune system to identify and neutralize antigens.¹ Each Ig molecule is composed of a basic structure that includes two identical heavy polypeptide chains and two identical light chains, which are linked together by multiple disulfide bonds.² These molecules are categorized into five classes (IgA, IgD, IgE, IgM, and IgG), which are differentiated by their heavy polypeptide chain type. IgG antibodies are particularly prominent in therapeutic applications due to their extended half-life.^{3–5} The increasing number of approved therapeutic antibodies highlights their growing importance in clinical practice, with over 200 marketed antibody therapeutics and nearly 1,400 investigational product candidates currently undergoing evaluation in clinical studies for the treatment of a wide variety of diseases.⁶ Among these molecules, monoclonal antibodies (mAbs) are engineered B-lymphocyte products designed for specific targeted antigen

binding, representing a prominent class of therapeutic antibodies.^{6,7} MAb, approximately 150 kDa in size, comprise two antigen-binding fragments and one Fc fragment.⁸ The Fc fragment can bind to various cell-associated receptors, including the neonatal Fc receptor (FcRn) and Fc gamma receptor.^{8,9} The interaction between mAbs and FcRn plays a pivotal role in the endosomal recycling of mAbs, thereby prolonging their lifespan within the organism.^{9,10} These molecular characteristics exert a substantial impact on the

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pharmacokinetic (PK) and pharmacodynamic (PD) attributes of mAbs, setting them apart from conventional small chemical molecules.⁸ Consequently, mAbs offer several therapeutic advantages, including an extended half-life spanning from 18 to 21 days,¹¹ enhanced potency, and improved selectivity with reduced off-target toxicity.

The utilization of PK modeling and simulation constitutes a pivotal practice in the preclinical development of mAbs to predict the first-in-human (FIH) dose, estimate activity, and anticipate the outcome of the planned clinical doses. Accurate PK predictions are essential not only for optimizing therapeutic efficacy but also for providing reliable data for safety studies. However, predicting human PK for therapeutic proteins, including mAbs, remains challenging due to the lack of standardized methods for mechanism-based elimination modeling. Physiologically based pharmacokinetic (PBPK) modeling has emerged as a powerful tool to address these challenges. PBPK models integrate physiological, biochemical, and drug-specific data to simulate drug disposition in the human body, offering a framework for predicting PK across populations and scenarios. Moreover, in the context of drug discovery and development, this approach enables the simulation of local target concentrations, thereby facilitating predictions of receptor occupancy and supporting early estimation of clinically effective doses. Common applications of PBPK modeling consist in guiding drug candidate selection,¹² drug–drug interaction (DDI) risk assessments,¹³ and prediction of PK in special populations.¹⁴ One common application is the prediction of human PK from preclinical data to support the design of FIH studies.¹³ PBPK modeling is increasingly recognized and encouraged by regulatory agencies in the approval of novel drugs.¹⁵ However, these PBPK models, predominantly designed for small molecules with well-characterized transporter and metabolic pathways, face limitations and challenges when applied to therapeutic proteins like mAbs. These challenges include accounting for the complex distribution and elimination pathways of mAbs, as well as their interactions with the FcRn.

The increasing use of PBPK modeling has been facilitated by the accessibility of PBPK software tools. In this study, all models were built using PK-Sim, an open-source PBPK modeling platform that is part of the Open Systems Pharmacology (OSP) suite.¹⁶ The hydrodynamic radius and the dissociation constant (K_d) for FcRn ($K_{d_{FcRn}}$) are key parameters for characterizing mAb PK in PK-Sim, as these are the main determinants for predictions of distribution and clearance, respectively. The hydrodynamic radius relates to the molecular size (approximated as a theoretical sphere), which in PK-Sim can be estimated from the effective molecular weight (MW).¹⁶ This estimation is based on an empirical relationship between MW and Stokes radius derived from published data on proteins of various sizes.¹⁶ This correlation assumes globular conformation and applies a power-law function to calculate the radius, which is then used to parametrize size-dependent processes such as extravasation and lymphatic transport within the two-pore formalism. In contrast, no reliable approach to inform $K_{d_{FcRn}}$ with high precision in PBPK modeling is currently available, making this parameter a major hurdle for accurate predictions of mAb PK. Although $K_{d_{FcRn}}$ can be experimentally determined and subsequently used as a PBPK model input, the high variability of these *in vitro* measurements, stemming from experimental inconsistencies, further complicates their practical application.^{17,18}

Common techniques used to measure $K_{d_{FcRn}}$ include micro-scale thermophoresis (MST), which relies on specific light-emitting diode and power settings. Variations in these settings or fluctuations in temperature during the assay can significantly impact the thermophoretic mobility and binding data.¹⁹ Surface plasmon resonance (SPR), which measures binding kinetics in real-time, is also sensitive to factors such as temperature and buffer composition.²⁰ Additionally, enzyme-linked immunosorbent assay (ELISA) quantifies binding affinity but differences in antibody coating, detection reagents, and incubation times impact the measurement.²¹ Each of these techniques, along with other methods, introduces variability that affects the accuracy and reliability of $K_{d_{FcRn}}$ measurement.²² Extrapolation from *in vivo* nonclinical data (e.g., monkeys, mice, rats) offer a potential solution, but limitations exist as previously discussed, when trying to identify the most reliable predictors of human $K_{d_{FcRn}}$ from preclinical data.²³ Selecting the appropriate species for determining $K_{d_{FcRn}}$ remains a complex task, as highlighted by the variability observed across different preclinical species (monkey, mouse, rat).²³ In the present work, we assessed whether a default reference value for the human $K_{d_{FcRn}}$ parameter, estimated from a reasonably large public domain database, could be used to predict accurately the human PK of new candidate IgG mAbs in the context of FIH trials.

METHODS

Database Creation. The selected molecules used in the study are listed in Table 1. Concentration–time profiles were digitized from publications and studies available in the public domain. The database was assembled using a combination of the following terms on PubMed and Google Scholar: pharmacokinetics, human (s), clinical trial, phase 1, phase 2, phase 3, and the name of mAbs. Plasma concentration–time profiles following intravenous (IV) administration were scanned using the tool WebPlotDigitizer. Based on a qualitative and visual assessment of the reported PK profiles for each mAb, concentration–time profiles exhibiting clear nonlinear characteristics indicative of a pronounced target-mediated drug disposition (TMDD) effect were excluded during database creation. Indeed, this approach assumes linear clearance to minimize confounding factors affecting elimination, with $K_{d_{FcRn}}$ considered the primary parameter influencing clearance.

The database, representing the global data set, was randomly divided into training and test data sets, using a 7:3 ratio of mAbs, for model development and evaluation.

PBPK Modeling. The IV plasma concentration data for each mAb were used to build a drug-specific PBPK model. This step was performed using the same structural distribution and elimination model for each drug assuming the absence of TMDD.¹⁶ In simulating or predicting PK profiles, the demographics of the study population of each study were matched with available information. Information on drug-specific MW was used when available; otherwise, a default value of 150 kDa was applied.⁷⁷ The PBPK software PK-Sim, part of the OSP suite version 11.1,⁷⁸ was used for all modeling activities.¹⁶ The mechanistic FcRn pattern implemented within the whole-body PBPK (WB-PBPK) model for large molecules was employed in this analysis.⁷⁸ Briefly, the WB-PBPK model, previously described by Niederalt et al.,¹⁶ consists of 15 compartments representing organs or tissues connected by blood and lymph flows. Each compartment is divided into

Table 1. Monoclonal Antibodies with Plasma Pharmacokinetic Data Following Intravenous Administration Evaluated in This Analysis (N = 50)

drug	M_w (kDa)	doses	refs
adalimumab	148	0.5, 1, 3, 5, 10 mg/kg	24
aducanumab	146	30, 60 mg/kg	25
atezolizumab	145	10, 20 mg/kg	26
belimumab	147	240 mg	27
bemarituzumab	150	6, 10, 15 mg/kg	28
benralizumab	150	1, 3 mg/kg	29
bevacizumab	149	3 mg/kg	30
bococizumab	145	12, 18 mg/kg	31
canakinumab	145	0.3, 1, 1.5, 3, 10 mg/kg	32
daclizumab	144	200, 400 mg	33
daratumumab	145	16 mg/kg	34
dupilumab	147	3 mg/kg	35
eculizumab	148	300 mg	36
emicizumab	146	0.25 mg/kg	37
enokizumab	148	1, 9 mg/kg	38
ensituximab	145	1.5, 2, 4 mg/kg	39
eptinezumab	150	100 mg	40
erenumab	150	140 mg	41
figitumumab	146	0.3 mg/kg	42
fresolimumab	144	1, 4 mg/kg	43
fulranumab	145	3, 10, 30 mg	44
gedivumab	147	3600, 8400 mg	45
gevokizumab	145	0.3, 1, 3 mg/kg	46
golimumab	150	100 mg	47
guselkumab	144	3, 10 mg/kg	48
infliximab	144	5 mg/kg	49
lecanemab	150	10 mg/kg	50
lesofavumab	145	1200, 3600, 8400 mg	51
mepolizumab	149	2.5, 10 mg/kg, 10, 75, 250, 750 mg	52,53
natalizumab	149	300 mg	54
obiltoximab	148	16 mg/kg	55
olokizumab	146	0.3, 1, 3, 10 mg/kg	56
omalizumab	149	150 mg	57
ozanezumab	145	1, 10, 15 mg/kg	58
pateclizumab	145	0.3, 3, 5 mg/kg	59
pertuzumab	148	420 mg	60
ralpancizumab	145	1, 3, 6 mg/kg	61
risankizumab	146	5 mg/kg	62
roledumab	149	3 mg	63
sasanlimab	146	1, 3, 10 mg/kg	64
secukinumab	151	3 mg/kg	65
sirukumab	150	100 mg	66
spesolimab	146	300 mg	67
tefibazumab	148	5, 10, 20 mg/kg	68,69
tezepelumab	147	210, 700 mg	70
tildrakizumab	147	0, 5, 3, 10 mg/kg	71
tislelizumab	144	2 mg/kg, 200 mg	72,73
tralokinumab	143	150 mg	57
trastuzumab	148	6 mg/kg	74
tremelimumab	149	6, 10, 16 mg/kg	75,76

vascular (plasma and blood cells), endosomal, interstitial, and cellular spaces. The transcapillary exchange of the drug between plasma and interstitial space by convection or diffusion in each organ is described using a two-pore formalism. The return of the drug from the interstitial space to the blood compartment is described by organ-specific lymph flow.¹⁶ It accounts for the disposition of mAbs in the vascular,

endothelial, and interstitial layers within each tissue compartment. Data processing, noncompartmental analysis (NCA), and graph plotting were performed in R version 4.2.2 coupled with RStudio version 2022.12.0.

Model Parameter Estimation. The $K_{d_{FcRn}}$ value was estimated by minimizing the residuals between the observed data and the model output. The estimation was conducted using PK-Sim with the Levenberg–Marquardt algorithm. The generic distribution model included in the large-molecule PBPK framework of PK-Sim was applied uniformly across all mAbs to ensure an unbiased comparison of estimated $K_{d_{FcRn}}$ values. $K_{d_{FcRn}}$ was the only parameter estimated for each mAb. Depending on the available data, one or more doses were used, and a single specific $K_{d_{FcRn}}$ was estimated for each mAb by simultaneously fitting all PK profiles across the different dosages.

Model Evaluation. For all of the simulations and predictions, model evaluation was conducted by visually comparing the PK profiles generated by the model with the observed data. Subsequently, using the NCA approach, predictive errors were calculated for the area under the curve from the time of the first to the last observed concentration (AUC), maximal plasma concentration (C_{max}), and half-life ($t_{1/2}$). The criteria for validating and verifying each model were derived from regulatory submission guidelines outlined by Abduljalil et al. and Shebley et al.^{79,80}

Identification and Predictive Performance of a Reference $K_{d_{FcRn}}$ Value. The training data set underwent a two-step process: first, model parameter identification and estimation of $K_{d_{FcRn}}$ for each mAb, followed by the determination of the training data set median $K_{d_{FcRn}}$ value. This value was then directly applied to predict the PK of mAbs in the test data set without further adjustment. In a subsequent step, $K_{d_{FcRn}}$ was also estimated for the mAbs in the test data set. This step was performed to enable evaluation of the global data set median $K_{d_{FcRn}}$ value and comparisons between predictions performed using the global data set median $K_{d_{FcRn}}$ and the best potential model performance (specific $K_{d_{FcRn}}$ for each mAb), leveraging all the available data.

RESULTS

Training Data Set. The final training data set included 35 mAbs. A comparison of PBPK simulations to observations for each of the 35 mAbs included in the training data set is displayed in Figure 1. For all the drugs in the training data set, after estimation of $K_{d_{FcRn}}$ for each mAb, the model-simulated AUC, C_{max} , and $t_{1/2}$ were within a 0.50–2.00 fold error compared to the observed values as illustrated in Figure 2. Moreover, the model-simulated AUC, C_{max} , and $t_{1/2}$ were within a 0.80–1.25 fold error compared to the observed values for 74% of the training data set.

The drug-specific $K_{d_{FcRn}}$ values of the 35 mAbs in the training data set were used to derive the training data set median $K_{d_{FcRn}}$ value, which was determined to be 1.05 μ M. Figure 3 shows the estimated $K_{d_{FcRn}}$ for each mAb, along with the distribution of these values. The maximum $K_{d_{FcRn}}$ value has been obtained for daratumumab, with a value of 3 μ M, and the minimum $K_{d_{FcRn}}$ has been obtained for spesolimab, with a value of 0.55 μ M.

Predictive Performance Assessment Using the Test Dataset. The final test data set included 15 mAbs. For each of these drugs, a PBPK model was developed using a fixed $K_{d_{FcRn}}$ value of 1.05 μ M, corresponding to the median value

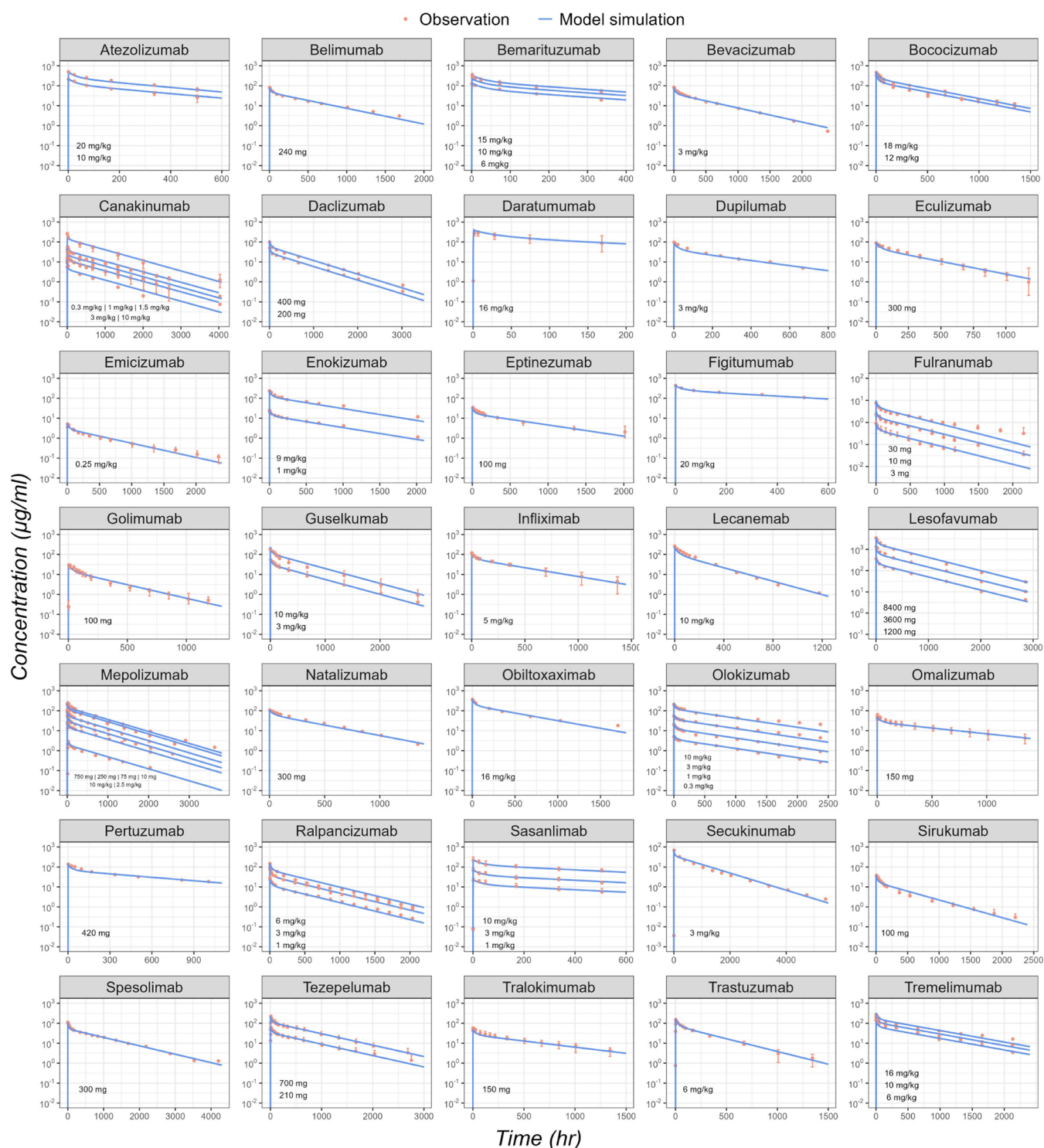


Figure 1. Simulated (lines) concentration versus time profiles after IV administration compared to observations (dots) for the 35 mAbs included in the training data set after the estimation of a drug-specific K_{dFcRn} .

previously determined from the training data set analysis. The comparison of observed versus predicted PK profiles for each molecule in the test data set is displayed in Figure 4. The predictive errors for AUC, C_{max}, and $t_{1/2}$ illustrated in Figure 5 show that 100% of the test data set was predicted within a 0.50–2.00 fold error range compared to the observed values. Furthermore, 53% of the test data set fell within the 0.80–1.25-fold error threshold.

In a subsequent step, drug-specific K_{dFcRn} was also estimated for each mAb in the test data set. Observed versus simulated PK profiles for each molecule in the test data set are displayed in Figure S3. For all the drugs in the test data set, simulations of AUC, C_{max}, and $t_{1/2}$ fell within a 0.50–2.00 fold error compared to the observed values as illustrated in Figure S4. Additionally, simulations of AUC, C_{max}, and $t_{1/2}$ fell within a 0.80–1.25 fold error compared to the observed values for 74% of the test data set.

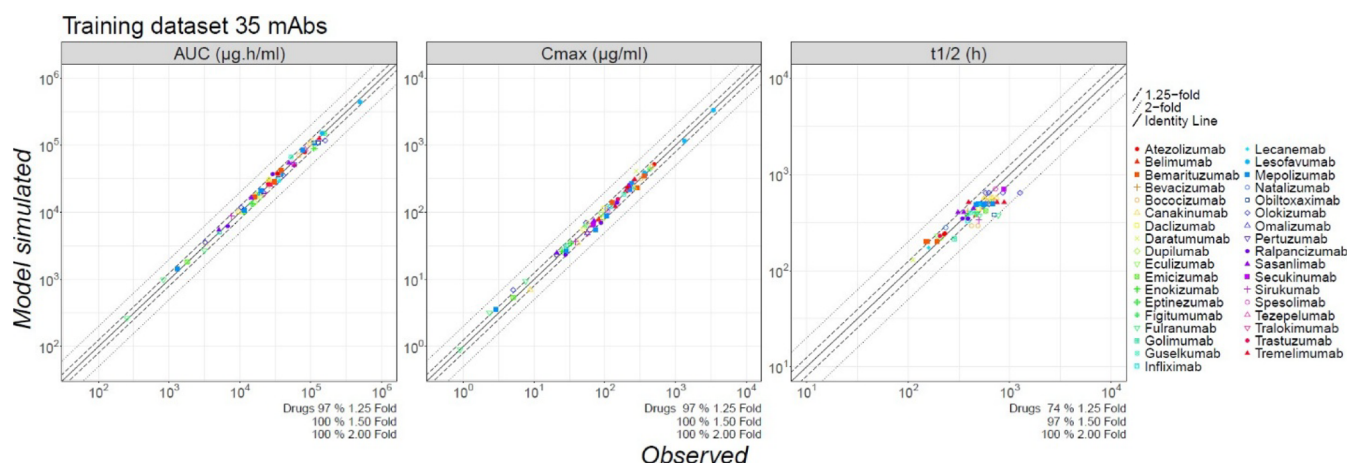


Figure 2. Simulated versus observed AUC, Cmax, and $t_{1/2}$ for the 35 mAbs included in the training data set after the estimation of a drug-specific $K_{d_{FcRn}}$.

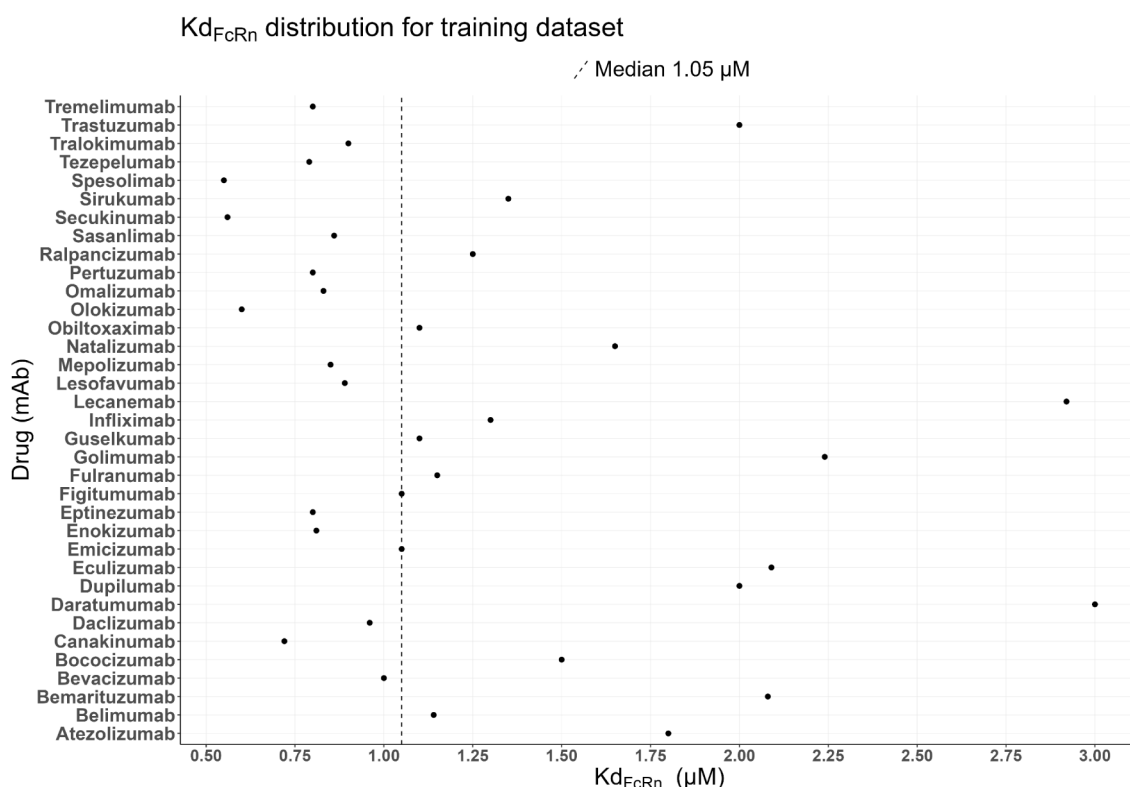


Figure 3. Estimated $K_{d_{FcRn}}$ values for the mAbs in the training data set. Individual drug-specific $K_{d_{FcRn}}$ estimates are shown as dots and the median as a dashed line.

Global Data Set. The individual drug-specific $K_{d_{FcRn}}$ values of the 50 mAbs (Table S1), combining training and test data sets, were used to derive a global data set median value, which was determined to be $1.07 \mu\text{M}$. Figure S1 illustrates the estimated drug-specific $K_{d_{FcRn}}$ for each mAb included in the global data set, along with the distribution of these values across both training and test data sets. The maximum and minimum $K_{d_{FcRn}}$ values were the same as those for the training data set.

Predictive Performance Assessment Using the Global Dataset. The $K_{d_{FcRn}}$ parameter value for each PBPK model developed for the drugs in the global data set was set to $1.07 \mu\text{M}$, i.e., the global median value of $K_{d_{FcRn}}$ determined from all

the available data (training and test data sets). Using this approach, a comparison of observed versus predicted PK profiles for each molecule is displayed in Figure 6. The predictive errors for AUC, Cmax, and $t_{1/2}$ illustrated in Figure 7 show that 92% of the global data set was predicted within a 0.50–2.00 fold error range compared to the observed values. Additionally, the predictions of AUC, Cmax, and $t_{1/2}$ fell within a 0.80–1.25 fold error range compared to the observed values for 48% of the global data set.

Comparison of Performance between the Global Reference $K_{d_{FcRn}}$ Value and Drug-Specific $K_{d_{FcRn}}$ Value. Comparison of the resulting PK parameters using estimated drug-specific $K_{d_{FcRn}}$ values versus the global $K_{d_{FcRn}}$ reference

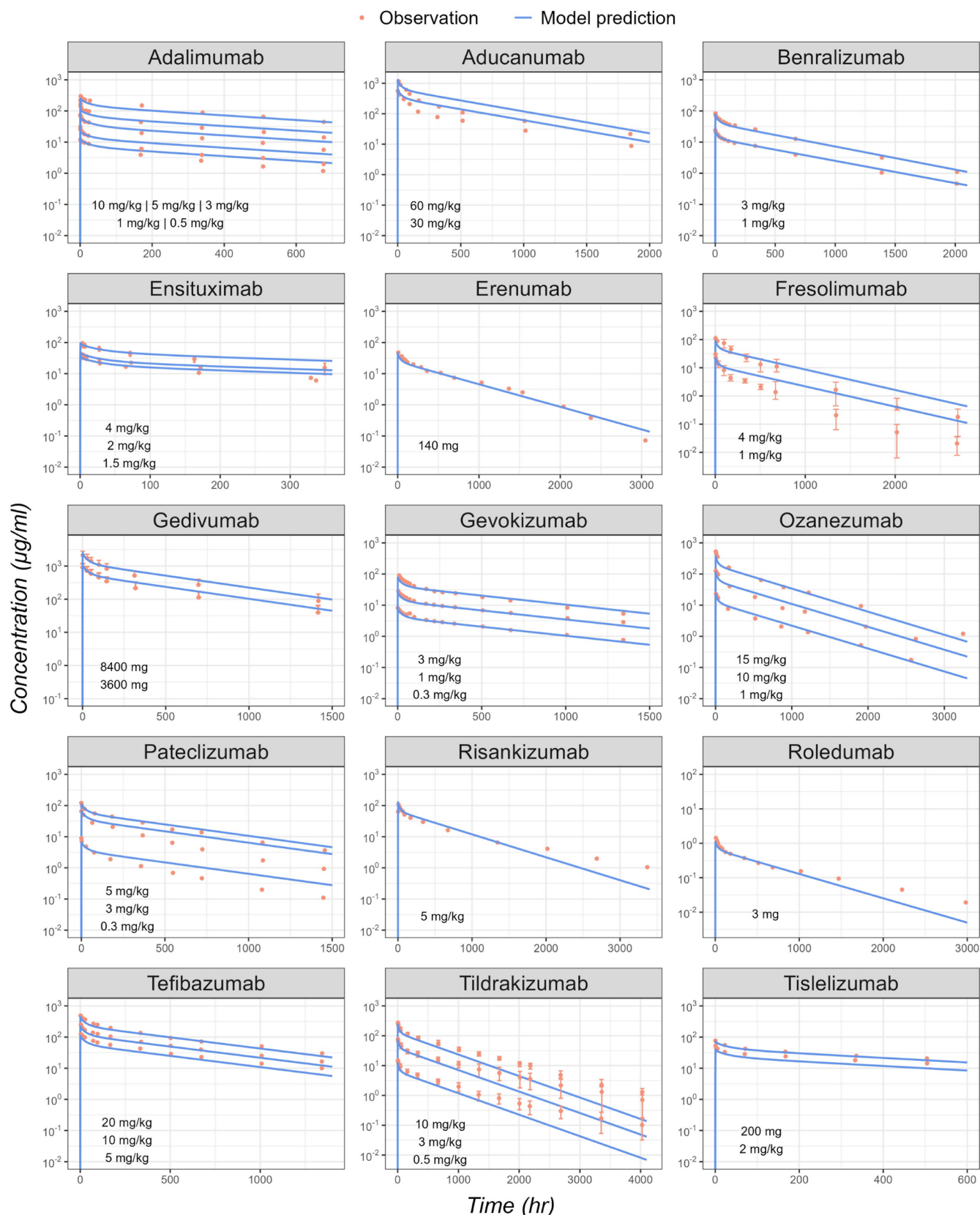


Figure 4. Predicted (lines) concentration versus time profiles after IV administration compared to observations (dots) for the 15 mAbs included in the test data set, using a $1.05 \mu\text{M}$ value for $K_{\text{D}_{\text{FcRn}}}$ determined from the training data set.

value was performed, i.e., a comparison between simulated (using drug-specific estimated $K_{\text{D}_{\text{FcRn}}}$) and predicted (using global reference $K_{\text{D}_{\text{FcRn}}}$) PK parameters. This was performed

for the 50 mAbs included in the global data set. Figure S5 illustrates the ratio (and the percent deviation from unity, i.e., 100%) between PK parameters predicted using the reference

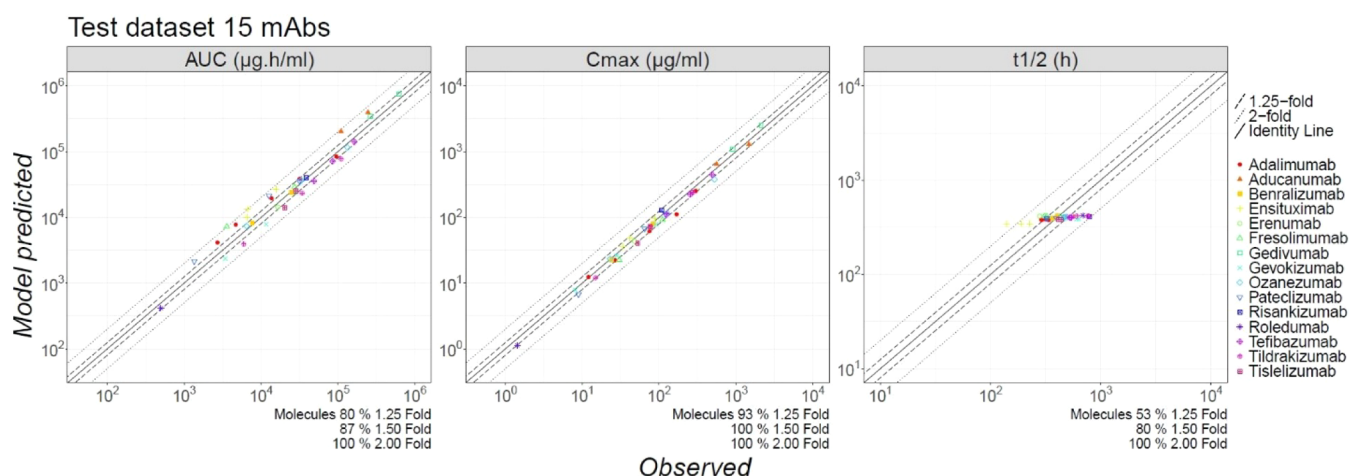


Figure 5. Observed versus PBPK model-predicted AUC, Cmax, and $t_{1/2}$ for the 15 mAbs included in the test data set, using a $1.05 \mu\text{M}$ value for $K_{d_{\text{FcRn}}}$ determined from the training data set.

$K_{d_{\text{FcRn}}}$ value of $1.07 \mu\text{M}$ and those predicted using the estimated drug-specific $K_{d_{\text{FcRn}}}$ value. For plasma exposure (AUC), 22 mAbs (44% of the global data set) and 49 mAbs (98%) displayed a deviation within the 0.80–1.25 fold error and within the 0.50–2.00 fold error, respectively. The maximum ratio was obtained for lecanemab (2.45). For $t_{1/2}$, 25 mAbs (50%) and 48 mAbs (96%) displayed a deviation within the 0.80–1.25 fold error and within the 0.50–2.00 fold error, respectively. The maximum ratio was obtained for daratumumab (2.22).

DISCUSSION

In this study, we analyzed a large set of mAbs clinical data via PBPK modeling with the goal of establishing a standardized approach for prospective predictions of mAb PK to inform FIH trials. A database of PK profiles was compiled for 50 intravenously administered mAbs. These molecules were selected based on the availability of published full time-course data, required to accurately characterize the elimination phase, and the absence of obvious TMDD characteristics. The ultimate outcome was to assess whether a generic propensity for endosomal recycling via the FcRn receptor could be used to adequately predict human PK of new candidate mAbs, in the context of FIH trials.

The approach was evaluated in two steps. First, we applied a training/testing data set approach in order to demonstrate the generalizability of the analysis. Among the 35 molecules included in the training data set: 14 were fully human, 19 were humanized, and 2 were chimeric. Furthermore, 23, 7, and 5 of the mAbs belonged to the IgG1, IgG2, and IgG4 subclasses, respectively. The test data set comprised 15 mAbs: 8 fully human, 6 humanized, and 1 chimeric. Of these, 11 were IgG1, 3 were IgG2, and 1 was IgG4. The distribution of these molecule specificities matched that of the training data set. Moreover, Figure S1 illustrates the homogeneous distribution of the estimated drug-specific $K_{d_{\text{FcRn}}}$ values for mAbs across both data sets. These findings confirm the similarity and balance between the two data sets. The training data set underwent a two-step process: first, model parameter estimation of $K_{d_{\text{FcRn}}}$ for each mAb, followed by the identification of the training data set median $K_{d_{\text{FcRn}}}$ value, which was determined to be $1.05 \mu\text{M}$. For comparison with *in vitro* measurements, endogenous IgG is reported to have a

$K_{d_{\text{FcRn}}}$ of $0.90 \mu\text{M}$ ^{19,81} using an MST assay, or ranging between 0.78 and $1.30 \mu\text{M}$ using an SPR assay at pH 6.^{82,83}

These findings indicate that the estimated value obtained with our approach falls within the reported experimental ranges of IgG affinity for FcRn. This value was then directly applied to predict the PK of mAbs in the test data set. Predictions were deemed accurate if the ratios between predicted and observed values fell within the 0.80–1.25 (bioequivalence) error range, acceptable if within 0.50–2.00 (2-fold) error range, and inaccurate if outside the 2-fold error range. We can observe the predictivity performance of this value in Figure 5, with AUC, Cmax, and $t_{1/2}$ falling within the acceptable 2-fold error range compared to the observed values for 100% of the test data set. Furthermore, 53% of the test data set falls within the bioequivalence range, demonstrating the generalizability of the approach.

In the second method, to evaluate the predictivity performance of a global data set median $K_{d_{\text{FcRn}}}$ value estimated using all the available data, the global data set underwent a two-step analysis. First, model parameter estimation of $K_{d_{\text{FcRn}}}$ and PK simulation were performed for each mAb in the test data set. The comparison of PBPK simulations to observations for each mAb of the test data set is displayed in Figure S3. Second, by combining the individual drug-specific $K_{d_{\text{FcRn}}}$ values from the training and test data sets, we identified the global data set median $K_{d_{\text{FcRn}}}$ value. This analysis yielded a median value of $1.07 \mu\text{M}$ and a mean of $1.24 \mu\text{M}$. The median and mean values closely matched the 1.05 and $1.27 \mu\text{M}$ values, respectively, obtained from our previously selected training data set. Moreover, the training data set contained the same maximum and minimum values compared to the global data set, with daratumumab showing a maximum value of $3 \mu\text{M}$, and spesolimab showing a minimum value of $0.55 \mu\text{M}$. This indicates that no significant bias was introduced during the creation of the respective data sets and that homogeneity was preserved throughout the data-splitting process. The global data set median $K_{d_{\text{FcRn}}}$ value of $1.07 \mu\text{M}$, estimated using all the available data, was designated as the reference $K_{d_{\text{FcRn}}}$ value and was then directly applied to predict the PK of all mAbs without further adjustment. We can observe the predictivity performance of the values, in Figure 7, with AUC, Cmax, and $t_{1/2}$ falling within the acceptable 2-fold error range compared to the observed values for 92% of the global data set, representing

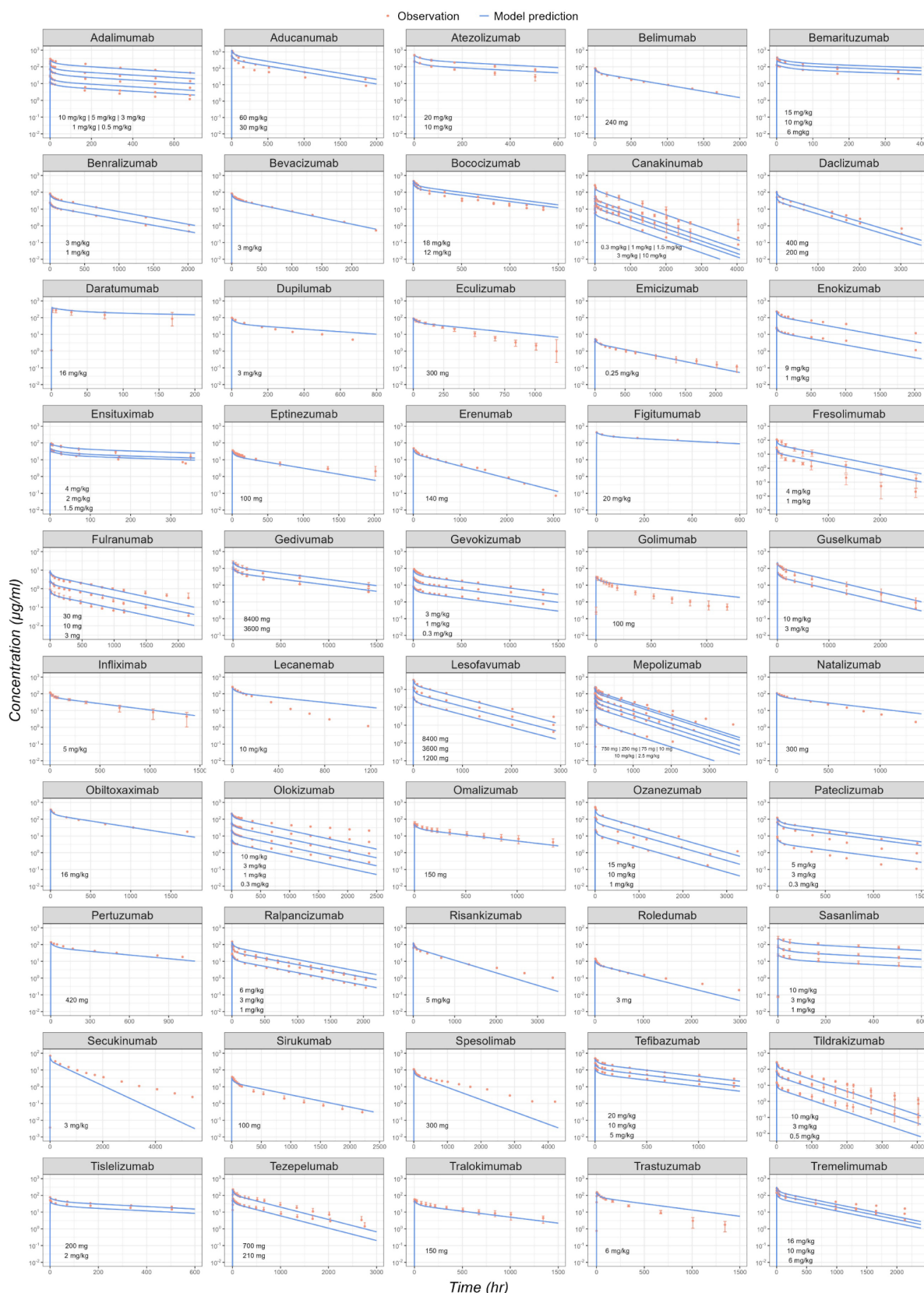


Figure 6. Predicted (lines) concentration versus time profiles after IV administration compared to observations (dots) for the 50 mAbs included in the global data set, using a $1.07 \mu\text{M}$ value for K_{dFcRn} determined from the global data set.

46 mAbs. Additionally, the predictions of AUC, C_{max} , and $t_{1/2}$ fell within the accurate bioequivalence range for 48% of the global data set, representing 24 mAbs.

When comparing the results of the two approaches, using either the reference K_{dFcRn} value (Figure 7) or the estimated individual drug-specific K_{dFcRn} value for each mAb (Figure S4),

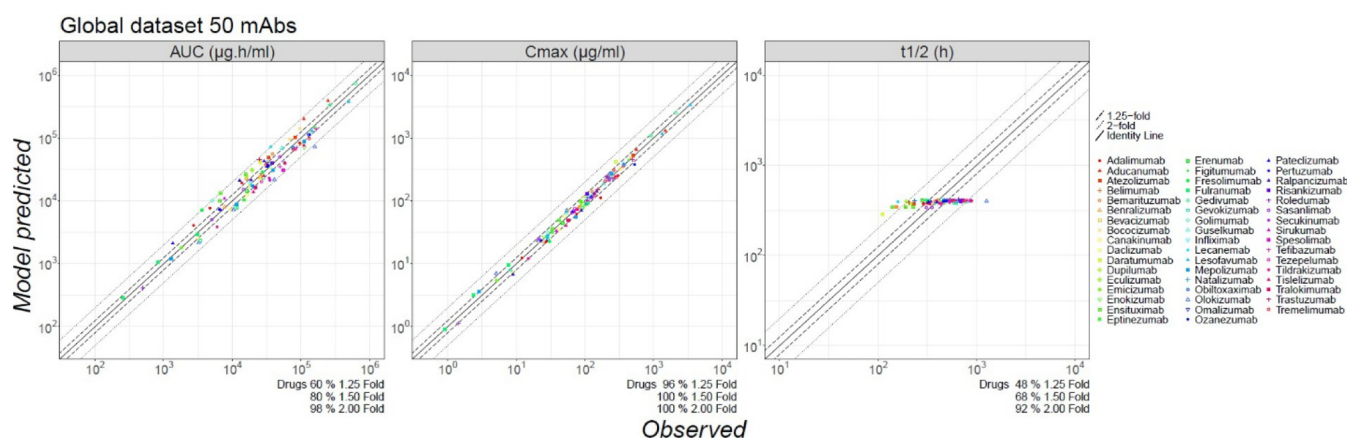


Figure 7. Observed versus PBPK model-predicted AUC, C_{max}, and *t*_{1/2} for the 50 mAbs included in the global data set, using a 1.07 μM value for K_d_{FCRN}, determined from the global data set.

we can observe that, following IV administration, the $K_{d_{FcRn}}$ parameter predominantly influences plasma exposure (AUC) and elimination ($t_{1/2}$) of mAbs, with no significant impact on C_{max} . Furthermore, to evaluate the sensitivity of AUC and $t_{1/2}$ parameters using either the reference $K_{d_{FcRn}}$ value or the individual drug-specific $K_{d_{FcRn}}$ value for each mAb (considered as the best potential model performance), we compared the values of AUC and $t_{1/2}$ for each mAb by calculating the ratio of the outcome result at each step. This comparison is illustrated in [Figure S5](#). The comparison for C_{max} is not included, as no impact was observed during the previous comparison of the two approaches. Except for daratumumab and lecanemab, the ratios of all other mAbs fell within the acceptable error range of 0.50–2.00 for AUC and $t_{1/2}$. [Figure S2](#) shows the overall distribution of $K_{d_{FcRn}}$ values across the global data set, which appears right-skewed, with most mAbs exhibiting values between 0.5 and 1.5 μM and a peak around 1.0 μM . Interestingly, these two molecules, daratumumab and lecanemab, were the only mAbs displaying markedly higher $K_{d_{FcRn}}$ values ($>2.5 \mu M$), suggesting weaker FcRn binding or possible involvement of additional clearance pathways beyond FcRn-mediated dynamics.

Lecanemab is a humanized IgG1 that selectively binds to soluble Amyloid- β aggregate species, including oligomers and protofibrils.⁵⁰ Fibrinogen has been identified as a major potential plasma protein binder for lecanemab,^{84,85} and it has been hypothesized that fibrinogen binding may influence lecanemab's bioavailability in clinically relevant ways, notably by acting as a reservoir and modifying its half-life.⁸⁴ Our analysis and PK studies of lecanemab reported a terminal half-life between 5 and 6.5 days (associated with a relatively high $K_{d_{FcRn}}$ value of 2.92 μM). This results in a faster elimination rate compared to typical IgG antibodies, generally exhibiting a half-life of 18 to 21 days.¹¹ Daratumumab is a human IgG1 monoclonal antibody targeting CD38, indicated for the treatment of multiple myeloma (MM).³⁴ In our analysis, the digitized data for this molecule comes from a study where daratumumab was administered to patients with relapsed or refractory MM.³⁴ Patients with MM exhibit high IgG levels (serum M-proteins produced by MM cells) that could compete with FcRn recycling and therefore interfere with daratumumab's FcRn recycling, resulting in a higher clearance. In this case, the observed disease interference might explain why daratumumab has a high estimated $K_{d_{FcRn}}$ value of 3 μM .

It is essential to recognize that the estimated Kd_{FcRn} values are directly influenced by the PK profiles used for the model fitting. Differences in study design, patient population, and dosing regimens can lead to variability in reported PK and, consequently, in Kd_{FcRn} estimation. For instance, alternative references report substantially longer half-lives for atezolizumab (~ 21 days⁸⁶ vs ~ 14 days²⁶ in the reference used in our analysis) and trastuzumab (~ 28.5 days⁸⁷ vs ~ 7 days⁷⁵ in the reference used in our analysis), which would result in lower Kd_{FcRn} estimates if those PK profiles were considered. Therefore, the conclusions drawn in this work may be specific to the analyzed mAbs and might not be broadly applicable to all of the mAbs. The generalizability of our findings largely depends on how representative the selected sample is. Although we aimed to include a diverse set of mAbs and maximize coverage, the study is limited by reliance on available literature sources, which may introduce bias. Nevertheless, the 50 mAbs included are believed to reasonably represent a large portion of the marketed antibodies with various indications. Variability across studies could not be fully captured in our approach, and future work should investigate this aspect to enhance the robustness and validity of the proposed workflow.

The proposed reference $K_{d_{FcRn}}$ value may also serve as an initial input in developing models for other mAbs with more complex PK properties, such as antibody-drug conjugates (ADCs), or other antibody-related formats with varying affinities to FcRn. It should also be noted that for mAbs presenting TMDD, it may be difficult to identify the different clearance pathways (TMDD and nonspecific clearance due to protein catabolism) if information is missing concerning the TMDD process. In this case, the use of the proposed $K_{d_{FcRn}}$ value could eventually support TMDD identification by fixing this parameter during other estimation processes. However, because our current framework does not explicitly account for TMDD, any residual clearance beyond FcRn-mediated recycling for mAbs is implicitly attributed to FcRn, which may lead to a slight overestimation of the $K_{d_{FcRn}}$ value in our study. This occurs because profiles selected for the apparent absence of TMDD may still include minor contributions from target-mediated clearance or other pathways that are not clearly characterized and are difficult to distinguish in seemingly linear PK profiles.⁸ Consequently, when extending a drug-specific PBPK model to include TMDD dynamics, the $K_{d_{FcRn}}$ value may tend to be slightly lower than the value

reported in our study. This limitation should be considered when interpreting $K_{d_{FcRn}}$ affinity estimates, and future work should aim to refine these estimates by using models that integrate TMDD and leverage data across multiple scenarios. Moreover, it is important to note that the reference $K_{d_{FcRn}}$ value estimated in this work might be specific to the PBPK modeling framework implemented in the open-source platform PK-Sim. Application of this value in other PBPK platforms would require careful adaptation of underlying assumptions and parameterization strategies.

In a previous work, we demonstrated, through regression equations with a data set of 27 mAbs, that extrapolating $K_{d_{FcRn}}$ values from animals to humans showed considerable limitations.²³ Compared to mice and monkeys, that work identified rats as the most reliable animal model, showing a strong correlation ($R^2 = 0.74$) between their $K_{d_{FcRn}}$ values and those of humans.²³ Using the generic PBPK model for large molecules in PK-Sim, one drug-specific parameter (i.e., MW) and the proposed reference value of $1.07 \mu\text{M}$ for $K_{d_{FcRn}}$, the observed data are well described for a majority of the literature-derived mAbs in the database. Our study indicates that the proposed reference $K_{d_{FcRn}}$ value as a default input in the large molecules model of the PK-Sim platform provides accurate predictions without additional input and that this constitutes a straightforward method for guiding FIH trials. This result is promising, as it highlights the potential to reduce reliance on animal testing in preclinical safety assessments. Consequently, this method offers a more efficient and ethical pathway for drug development.⁸⁸ In response to the limitations and ethical concerns associated with animal testing, the U.S. Food and Drug Administration (FDA) has recently emphasized the need to reduce animal use in the evaluation of mAbs. The FDA highlighted the need for more ethical, innovative, and efficient alternatives such as quantitative modeling solutions, including PBPK modeling, which can provide accurate and predictive insights.⁸⁹ In line with this expectation, the present work proposes a valuable approach to reduce the reliance on animal studies.

CONCLUSION

This study presents the PBPK models of 50 clinical mAbs developed using the open-source PBPK platform PK-Sim and a robust approach for naïve predictions of human mAbs linear PK. The presented approach requires only the drug MW as input, and the PBPK generic model for large molecules in PK-Sim, to predict the human PK of mAbs (IgGs) accurately in 48% of cases and acceptably well in 92% of cases. Based on the demonstrated predictive performance, it establishes a benchmark for predicting the linear PK of mAbs in humans (excluding scenarios with pronounced TMDD) and provides an efficient alternative to traditional preclinical models, which often exhibit limited predictivity, in guiding FIH trials in the absence of human PK data. Further external validation is warranted, and future work should include verification with newly developed mAbs to enhance the data set's biophysical space and relevance.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscptsci.5c00674>.

Estimated drug-specific (individual) $K_{d_{FcRn}}$ values for all the 50 mAbs; distribution and median of estimated drug-specific $K_{d_{FcRn}}$ values across all 50 mAbs; distribution of $K_{d_{FcRn}}$ values across global data set; comparison of model-simulated PK profiles with observed data for the 15 mAbs of the test data set after IV administration; observed versus PBPK model-simulated AUC, C_{max} , and $t_{1/2}$ for all the 50 mAbs included in the global data set; predicted versus simulated AUC and $t_{1/2}$ for the 50 mAbs included in the global data set, using reference $K_{d_{FcRn}}$ value of $1.07 \mu\text{M}$ versus individual drug-specific $K_{d_{FcRn}}$ value for each mAb (PDF)

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Author Contributions

Data analysis: S.B.; investigation: S.B. and D.T.; project supervision: D.T., F.G., E.S., and L.N.; writing original draft: S.B.; writing, review, and editing: S.B., D.T., F.G., E.S., L.N., A.D., and M.C. All authors have read and agreed to the published version of the manuscript.

Notes

The authors declare the following competing financial interest(s): SB, AD, LN and DT are Sanofi employees and may hold shares and/or stock options in the company. ES and DT are members of the OSP Management Team. FG and MC have nothing to disclose.

ABBREVIATIONS

ADC antibody-drug conjugate

AUC	area under the curve
C _{max}	maximum concentration
DDI	drug–drug interaction
Fc	crystallizable fragment
FcRn	neonatal fragment crystallizable receptor
FDA	Food and Drug Administration
FIH	first-in-human
IgA	immunoglobulins A
IgD	immunoglobulins D
IgE	immunoglobulins E
IgG	immunoglobulins G
IgM	immunoglobulins M
Igs	immunoglobulins
IV	intravenous
K _{d_{FcRn}}	dissociation constant for FcRn
mAb	monoclonal antibody
mAbs	monoclonal antibodies
MW	molecular weight
NCA	noncompartmental analysis
OSP	Open Systems Pharmacology
PBPK	physiologically based pharmacokinetic
PD	pharmacodynamic
PK	pharmacokinetics
<i>t</i> _{1/2}	half-life
T _{max}	time to reach maximum concentration
TMDD	target-mediated drug disposition

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